CONSERVATIVE CFD MODELLING OF IODINE DEPOSITION IN THERMAL-GRADIENT COLUMNS: NUMERICAL VERIFICATION AND INPUT SENSITIVITY

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ABSTRACT

Interpreting iodine deposition in thermal-gradient columns requires a consistent treatment of transport, gas speciation and exchange with condensed wall deposits. We present an OpenFOAM finite-volume model for nine Pb–Bi–I gases in a frozen helium carrier, with element-conserving gas equilibrium and shared wall inventories. Reversible iodide exchange follows a Hertz–Knudsen–Schrage law in series with the diffusive resistance between a cell centre and its wall face. Published thermosublimatography measurements supply three temperature profiles and deposition scans; unreported source histories and flow-reference conditions are treated as assumptions. Fifteen baseline and 64 follow-up simulations separate numerical refinement from selected input sensitivities. The baselines use bounded limited-linear advection on 41,472 cells; the follow-up sensitivities use first-order upwind advection on 18,432 cells and are reported as differences from their own baselines. In 60 s runs, the largest raw element-balance residue is 1.41×10^{-12} of supplied material. Halving the time step changes iodine deposition profiles by at most 0.323%, and eight instead of three species correctors by at most 0.021%. In the most mesh-sensitive case, upwind advection differs from the bounded scheme by 7.7–20.8% at equal mesh, and the bounded-scheme profile changes by 1.98% from 41,472 to 93,312 cells; apparent orders are inconsistent, so no formal grid-convergence index is claimed. A $+29\,\mathrm{kJ\,mol^{-1}}$ shift in the BiI gas Gibbs energy changes the BiI_3 share of deposited iodide iodine in the steel case from 94.58% to 2.01%, a strong but conditional thermochemical sensitivity. For silica, two shifts derived from discrepancies between the published temperature curve and its peak labels reduce 4–5 cm position errors to 0–1 cm, but they do not establish a corrected experimental coordinate system. Without tuning, the same wall law reproduces the deposition-onset temperatures of five independent single-species PbI_2 experiments with an RMS deviation of 18 K, against 35 K for both models of the study that reported them, although the computed deposit peaks lie 1–4 cm upstream. The results support a conservative computational method and an explicit assessment of uncertain inputs; they neither establish independent predictive validation nor reproduce the three-hour experiment and subsequent cooling.

Keywords Iodine deposition · Thermosublimatography · OpenFOAM · Gas equilibrium · Element conservation · Input sensitivity

1 Introduction

Iodine released from lead–bismuth eutectic (LBE) can travel in several chemical forms before it deposits on cooler surfaces. The chemical form matters because it sets both the gas-phase inventory available for transport and the equilibrium pressure that governs wall exchange. Liu et al. [1] studied this behaviour by evaporating pure iodides and iodine-containing LBE into helium, separating the products along a negative temperature gradient and characterising the deposits. Their measurements give spatially resolved evidence for PbI_2 and BiI_3 deposition, and for potassium-containing deposits in some LBE experiments. Linking such measurements to a transport calculation needs more than

a single deposition temperature: the temperature-coordinate mapping, the carrier flow, the source history and the gas composition all affect the predicted distribution.

Seen as a fluid-dynamics problem, a thermal-gradient column is a slow laminar flow of an inert carrier through a long, narrow tube whose wall temperature falls by several hundred kelvin along its length (fig. 1). Trace vapours released near the hot end are advected downstream by a carrier whose density and velocity vary with the imposed temperature, and they diffuse radially towards the wall. Where the local vapour pressure exceeds the equilibrium pressure at the wall temperature, material is lost to the wall; where it falls below that pressure, existing deposits can evaporate again. The deposition profile is therefore set by the competition between axial advection, radial diffusion to the wall and the interfacial exchange law. This is a Graetz-type wall mass-transfer problem [2], but with a nonlinear wall condition that depends on the local wall temperature and condensed inventory. The same coupling underlies the interpretation of thermochromatographic experiments [3–5].

Lobresco et al. [6] compared temperature-based and vapour-pressure-driven CFD deposition models. Temperature-based models require a calibrated deposition law; vapour-pressure-driven models link deposition to vapour-pressure data through interfacial kinetics. Their study also motivates extending transport models to multiple chemical species and to thermodynamic equilibrium. Its five PbI_2 experiments, with measured deposition temperatures at three flows and three durations, also provide a single-species test of the transport and wall-exchange model used here. A multispecies model adds a consistency requirement. Changes in chemical form must conserve the transported elements, and different gas pathways that lead to the same condensed material must share one physical wall inventory. Even an internally consistent implementation can give uncertain experimental predictions if its thermodynamic or source inputs are poorly constrained.

We address these issues in the finite-volume framework of OpenFOAM [7]. The model transports nine Pb–Bi–I gases in a prescribed helium carrier. Homogeneous gas equilibrium redistributes the local elemental inventory, and separate interfacial channels exchange material with immobile condensed wall inventories. The wall flux combines interfacial kinetic conductance with diffusion over the distance from the neighbouring cell centre to the wall. This avoids representing deposition through an arbitrarily chosen volumetric wall-layer thickness. It does not remove spatial discretisation error.

General thermodynamic packages such as GEM-Selektor and its GEMS3K kernel [8] also support equilibrium-based reactive transport. The software developed here has a GEMS3K interface, but the calculations reported in this paper use a specialised ideal-gas element-potential solver with tabulated formation constants. We therefore distinguish the thermodynamic formulation from the numerical engine used for the reported simulations. Condensed phases enter through wall exchange, not through homogeneous cell equilibrium.

All experimental input comes from the published paper. No author-supplied temperature files, release histories or mass-flow-controller reference conditions are available. We therefore do not infer a unique physical source from the deposition data. Instead, we report the consequences of two explicit source models and of selected variations in flow, diffusion, interfacial accommodation and thermodynamic data. Separate time-step and mesh studies show which conclusions survive numerical refinement. We treat a temperature-coordinate discrepancy in the silica experiment as an input-consistency problem: alternative mappings are defined from published quantities before the resulting CFD profiles are compared.

The contribution is a documented, conservative coupling of transport, gas equilibrium and wall exchange, together with a completed numerical and input-sensitivity assessment. The evidence comprises 15 baseline and 64 follow-up simulations of the three available experimental configurations, production re-runs of the baselines with a bounded advection scheme on a finer mesh, a scheme and mesh study up to 93,312 cells and a full-run iteration check. These are computational cases, not independent experimental replicates. In addition, the same model is compared, without adjustment, with five single-species PbI_2 experiments of Lobresco et al. and with the two models of that study. The scope is deliberately limited to the executed model: a fixed carrier and temperature field, prescribed release, ideal-gas speciation, and wall deposits without aerosol formation or surface-material-specific chemistry. Comparisons with experiment are conditional assessments of this model, not a claim of independent predictive validation.

2 Experimental basis and input reconstruction

This section summarises the published experiments and explains how the model inputs were reconstructed from them.

Table 1: Published configurations used in the baseline matrix. Sample masses are initial masses, not the amounts released in a 60 s run. Every experimental evaporation period was 10,800 s.

Identifier	Column	Diameter (mm)	Sample (mg)	He (mL/min)
PbI2_SS	Steel	4.8	14.6	100
LBE-I_SS_II	Steel	4.8	346.0	45
LBE-I_SiO2_I	Silica	5.0	368.3	45

2.1 Available configurations

Liu et al. [1] used 1.25 m columns with an inner diameter of 4.8 mm for 316L stainless steel and 5.0 mm for fused silica. A 15 mm sample boat was inserted into a hot starting furnace, followed by a second furnace that imposed a decreasing temperature profile. Helium carried the evaporated material downstream. After three hours the source was withdrawn, the furnaces and gas flow were turned off, and the column cooled before the deposits were analysed. Temperature measurements and radiotracer scans used 1 cm increments.

Table 1 lists the three configurations for which both temperature curves and iodine scans could be reconstructed. The stand-alone BiI_3 experiment at 800 °C was considered during case preparation but was not run, because the supplied figures do not include its axial temperature profile. This case is neither given a surrogate profile nor counted as an unsuccessful simulation.

Both LBE samples had a reported iodine mole fraction of 9.04×10^{-4} and a 900 °C starting furnace. Their reported deposited iodine amounts were 0.191 and 0.203 mg, respectively. These amounts set the prescribed mean iodine source in the present cases. They are not direct measurements of a time-resolved evaporation rate, nor an independent measurement of inlet speciation.

2.2 Digitisation and coordinate assumptions

We digitised the axial temperature curves and iodine histograms from the published figures. Temperatures are interpolated linearly to cell centres and wall faces; beyond the plotted interval they are held at the nearest endpoint value. The computational coordinate is taken to be the plotted coordinate. Its origin relative to the physical boat and furnace has not been independently confirmed. Figure 1 shows the reconstructed curves. The original baseline matrix keeps these assumptions; the silica-coordinate study varies them explicitly.

The recorded digitisation resolutions are approximately 1 mm in position, 4 K in temperature and 0.4 percentage points in histogram height. They describe raster extraction only, not the full experimental uncertainty: coordinate registration, detector response, occluded curve segments, temperature reproducibility and calibration are not included. Likewise, the uncertainties printed with the experimental deposition temperatures must not be read as uncertainties on every digitised profile bin. The reconstructed histograms sum to 99.17%, 96.70% and 102.24% for pure PbI_2, steel LBE and silica LBE. The primary comparison keeps these sums and does not silently renormalise them.

In the LBE scans, the paper assigns 6.6% and 7.0% of the iodine to KI, which lies outside the Pb–Bi–I species set. We scale the model profile to the remaining 93.4% or 93.0%, while the measured histogram keeps all its published deposits. A second, explicitly conditional comparison normalises both profiles over their common scanned range (section 6). Neither comparison supplies the omitted potassium chemistry.

2.3 Simulation duration and measurement protocol

All source-driven cases run for 60 s with a source rate based on the 10,800 s experimental period. They therefore release $1/180$ of the assumed three-hour inventory; the experimental inventory is not compressed into a one-minute calculation. With a stationary continuous source, the results can inform deposition rates and conditional profile shapes, but the cumulative wall inventories still contain the initial filling transient. Four additional cases stop the source and the carrier flow at 60 s and continue to 80 s with the original temperature profile. They probe the relaxation of residual gas, not the experimental cooling history (section 4).

3 Transport, speciation and wall exchange

This section states the model: trace-species transport in a frozen carrier, homogeneous gas equilibrium, reversible and reactive wall exchange, and the two prescribed source models.

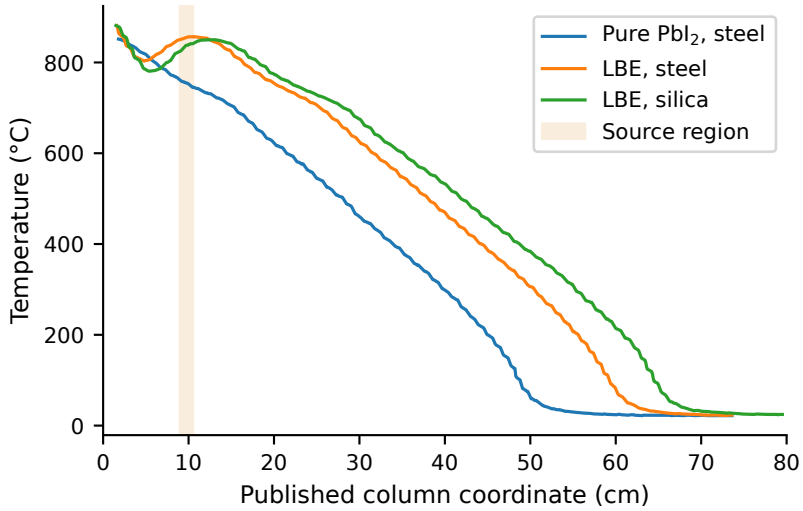


Figure 1: Axial temperature curves digitised from Liu et al. [1] for the three configurations, with the assumed computational source interval. Beyond the plotted range, the CFD domain holds the endpoint values constant. The curves reconstruct published figures; they are not temperature fields predicted by a conjugate heat-transfer calculation.

3.1 Frozen carrier and trace-species equations

The carrier fields ρ, $\mathbf{u}$, T and p come from a steady laminar helium calculation and are then held fixed. The reconstructed axial temperature is imposed over each whole cross-section. A gas species i is represented by Y_i, its mass per unit carrier mass, and satisfies

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + S_i^{\mathrm{rel}} + S_i^{\mathrm{wall}}. \tag{1}$$

Gas equilibrium is applied as a local redistribution between transport steps. The species do not alter the carrier density, momentum or temperature. The transported gases are Pb, PbI, PbI_2, Bi, Bi_2, BiI, BiI_3, I and I_2; helium is inert. Condensed PbI_2, BiI_3, Pb and Bi are stored as immobile wall inventories. In the code, solid-state identifiers label these inventories; the equilibrium-pressure tables select the crystalline or liquid state where applicable.

The model does not represent bulk nucleation, aerosols and their thermophoretic deposition [9, 10], thermal (Soret) diffusion [11, 12], hydrogen/oxygen chemistry, latent-heat feedback or chemical reaction with the steel or silica substrate. Differences between the column cases therefore come from their prescribed thermal profiles, dimensions and source inputs, not from a fitted substrate-specific sticking law.

3.2 Homogeneous ideal-gas equilibrium

Let ν_{ki} be the number of atoms of element $k \in \{\mathrm{Pb}, \mathrm{Bi}, \mathrm{I}\}$ in gas i. At fixed local temperature the transported element concentrations are

$$b_k = \sum_i \nu_{ki} \frac{\rho Y_i}{M_i}. \tag{2}$$

The equilibrium solver finds the gas partial pressures and dimensionless element potentials λ_k that satisfy

$$\begin{aligned} \frac{p_i}{p^\circ} &= K_{f,i}(T) \exp\left(\sum_k \nu_{ki} \lambda_k \right), \\ \sum_i \nu_{ki} \frac{p_i}{RT} &= b_k, \qquad p^\circ = 10^5\ \mathrm{Pa}. \end{aligned} \tag{3}$$

The tabulated $K_{f,i}$ share a common elemental reference and the stated standard pressure. Concentrations are converted back through $Y_i = M_i p_i / (\rho R T)$. Re-speciation therefore conserves elements, not individual species. The helium pressure is not an additional reactive element constraint; under the trace-species approximation its field stays prescribed.

The solver works in logarithmic variables with a safeguarded Newton iteration. The Pb and Bi subproblems are reduced at fixed iodine potential, and a Schur complement treats the remaining coupling. Canonical initialisation makes a restarted case independent of an unsaved equilibrium warm state. Re-speciation is requested at every time step in cells with $T \geq 350\,\mathrm{K}$; below this threshold the transported gas composition is kept. This cutoff is a numerical modelling choice, not a demonstration that reactions cease at 350 K. Homogeneous condensed phases are excluded, and instantaneous gas equilibrium stands in for unmeasured finite reaction rates.

3.3 Reversible iodide exchange

For PbI_2 and BiI_3, the kinetic wall law defines the flux as positive from condensate to gas:

$$j_i = G_i\,[p_{\mathrm{eq},i}(T_w) - p_{i,w}], \qquad G_i = C_e s_i \frac{2\alpha_i}{2-\alpha_i}\sqrt{\frac{M_i}{2\pi R T_w}}. \tag{4}$$

Here j_i is in $\mathrm{kg\,m^{-2}\,s^{-1}}$, M_i is in $\mathrm{kg\,mol^{-1}}$, pressures are in Pa, α_i is the accommodation coefficient and C_e, s_i are dimensionless multipliers. The baseline uses $C_e = s_i = \alpha_i = 1$. This is the Hertz–Knudsen–Schrage (HKS) form [13–15] used in vapour-pressure-driven deposition modelling [6]. The condensed-state pressure follows from the gas and condensed Gibbs energies on the stable tabulated condensed branch. The wall temperature selects this pressure; no deposition threshold is prescribed.

The kinetic law must also connect to the cell-centre concentration through molecular diffusion. With d_f the cell-centre-to-wall distance, $p_{i,c} = \chi_{i,c} Y_{i,c}$ and $\chi_{i,c} = \rho_c R T_c / M_i$, eliminating the wall concentration gives

$$\begin{aligned} G^{\mathrm{eff}}_{i,f} &= \frac{G_i}{1 + G_i \chi_{i,c} d_f / (\rho D_i)_f}, \\ j_{i,f} &= G^{\mathrm{eff}}_{i,f}[p_{\mathrm{eq},i}(T_w) - \chi_{i,c} Y_{i,c}]. \end{aligned} \tag{5}$$

This places the kinetic and half-cell diffusion resistances in series. The gas equation receives the face flow $A_f j_{i,f}$ as an implicit cell contribution, and the wall inventory receives its negative. Evaporation is limited by the face inventory available over the time step, so a bare wall cannot evaporate. Inventory updates use the exchange actually realised by the final gas solve, including active bounds, not an independently evaluated explicit predictor. Switching between deposition and evaporation therefore does not create material.

3.4 Shared metal inventories and a reactive channel

Additional wall channels connect Pb(g) to condensed Pb, and Bi(g) and Bi_2(g) to one common condensed Bi inventory. Each channel has its own equilibrium pressure and stoichiometric mass conversion, and all use the baseline interfacial coefficients. The reactive channel

$$3\mathrm{BiI}(g) \longrightarrow 2\mathrm{Bi(condensed)} + \mathrm{BiI_3}(g) \tag{6}$$

is enabled only in the forward direction in the executed cases. For unit activity of condensed Bi, the dimensionless reaction constant and effective reactant equilibrium pressure are

$$K_r(T_w) = \exp\left[-\frac{2G^\circ_{\mathrm{Bi,c}} + G^\circ_{\mathrm{BiI_3,g}} - 3G^\circ_{\mathrm{BiI,g}}}{RT_w}\right], \tag{7}$$

$$p_{\mathrm{eq,BiI}} = p^\circ \left[\frac{p_{\mathrm{BiI_3,c}}/p^\circ}{K_r(T_w)}\right]^{1/3}, \tag{8}$$

$$j_{\mathrm{BiI}} = \min\{0,\, G^{\mathrm{eff}}_{\mathrm{BiI}}[p_{\mathrm{eq,BiI}} - p_{\mathrm{BiI,c}}]\}. \tag{9}$$

All Gibbs energies in K_r are evaluated at T_w. This channel is a phenomenological treatment: it applies the HKS conductance to the reaction driving force, with the same half-cell resistance as the other channels. The coproduct pressure is taken from the beginning-of-step gas field, and the minimum suppresses the reverse reaction. The reaction consumes three BiI molecules, returns one BiI_3 molecule to the gas and adds two Bi atoms to the shared wall inventory. All channels draw on this one Bi inventory; a separate inventory for each reaction would duplicate a physical material. Coproducts are applied after the transported reactants have been solved. This irreversible channel is a model assumption and must not be read as a fully reversible surface-equilibrium model.

3.5 Prescribed source models

Source model Q1 releases PbI_2 and, for LBE, BiI_3 at constant rates

$$q_{1,i} = \frac{m_i}{M_i \tau_{\mathrm{exp}}}, \qquad \tau_{\mathrm{exp}} = 10{,}800\,\mathrm{s}. \tag{10}$$

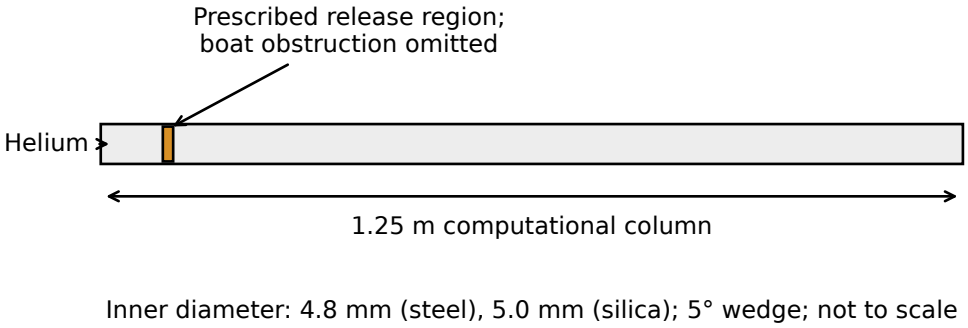


Figure 2: Computational geometry and prescribed source region. The source is distributed over selected cells; the sketch does not show a resolved boat surface. The model keeps the tube length and the material-specific diameters, and the 5° wedge provides the axisymmetric reduction.

For pure PbI_2, $m_i = 14.6\,\mathrm{mg}$ is the initial sample mass, and complete uniform release over the experimental period is assumed. For steel LBE, $m_{\mathrm{PbI_2}} = 0.111\,\mathrm{mg}$ and $m_{\mathrm{BiI_3}} = 0.181\,\mathrm{mg}$; for silica LBE, they are 0.243 and 0.085 mg. These compound masses were inferred from the deposits reported in the experimental paper. Q1 therefore conditions both the iodine loading and much of the species split on the measurements.

Source model Q2 instead supplies atomic Pb, Bi and I to the gas-equilibrium system. With q_{He} the full-column helium molar flow and p_h the source pressure, its elemental rates are

$$q_{2,\mathrm{Pb}} = q_{1,\mathrm{PbI_2}} + q_{\mathrm{He}}\beta p_{\mathrm{Pb}}(T_h)/p_h, \tag{11}$$

$$q_{2,\mathrm{Bi}} = q_{1,\mathrm{BiI_3}} + q_{\mathrm{He}}\beta[p_{\mathrm{Bi}}(T_h) + 2p_{\mathrm{Bi_2}}(T_h)]/p_h, \tag{12}$$

$$q_{2,\mathrm{I}} = 2q_{1,\mathrm{PbI_2}} + 3q_{1,\mathrm{BiI_3}}. \tag{13}$$

The metal pressures use ideal LBE activities $a_{\mathrm{Pb}} = 0.447$ and $a_{\mathrm{Bi}} = 0.553$ at $T_h = 1173.15\,\mathrm{K}$. The multiplicative metal saturation β is unmeasured. Q2 changes the metal supply and lets the iodide split emerge from the model, but its iodine source is still fixed by the deposited iodine. Q2 is therefore a source hypothesis, not an independent identification of LBE evaporation. The rates are distributed over an assumed source region, without resolving a liquid surface, depletion of the physical boat or the boat's obstruction of the flow.

4 Numerical implementation and study design

This section describes the geometry and carrier preparation, the discretisation and coupling, and the completed case matrix.

4.1 Geometry and carrier preparation

A structured axisymmetric 5° wedge represents the 1.25 m column (fig. 2). The steel and silica radii are 2.4 and 2.5 mm. The production mesh has $576 \times 72 \times 1 = 41{,}472$ cells. The 64 follow-up sensitivities use the original $384 \times 48 \times 1 = 18{,}432$-cell mesh. The mesh studies use $256 \times 32 \times 1 = 8{,}192$, 18,432, 41,472 and, for one case, $864 \times 108 \times 1 = 93{,}312$ cells; successive levels refine the axial and radial spacing by a factor of 1.5. The source occupies $x = 0.09$–0.105 m and the radial interval $r = 0.25R$–$0.75R$. Its position and cell selection are computational assumptions, and the physical boat is omitted from the carrier geometry. Full-column source and inlet rates are multiplied by $5/360 = 1/72$ for the wedge.

The inlet flow rates are converted to helium mass flow assuming a controller reference of 298.15 K and 101,325 Pa. The supplied paper does not specify these reference conditions. Full-column rates are $2.727 \times 10^{-7}\,\mathrm{kg\,s^{-1}}$ at 100 mL/min and $1.227 \times 10^{-7}\,\mathrm{kg\,s^{-1}}$ at 45 mL/min. A separate study uses 273.15 K at the same reference pressure. Helium follows the perfect-gas law with constant heat capacity and Sutherland viscosity (appendix A). Steady laminar carriers are computed with `rhoSimpleFoam`, while the reconstructed temperature profile stays prescribed.

The converged face mass flux is kept, rather than reconstructed from interpolated velocity. At these low mass flows, a zero-initialised potential correction removes residual discrete divergence before species transport. Carrier preparation rejects a carrier if its original cell or global flux defect exceeds 10^{-6} of the inlet flow, or if a flux or velocity correction

exceeds 10^{-5} of the corresponding scale. After correction, inlet, global and cell continuity and leakage through impermeable boundaries are checked against 10^{-12} of the inlet flow. This procedure makes the frozen transport field discretely conservative. It is neither a new physical flow model nor evidence that the velocity field has converged.

The stated inputs fix the transport regime. With the full-column helium mass flow $\dot{m}$, the tube diameter d and the Sutherland viscosity of appendix A, $\mathrm{Re} = 4\dot{m}/(\pi d\mu)$ is 1.5–3.0 for the 100 mL/min steel case and 0.7–1.3 for the 45 mL/min cases between the source-furnace temperature of 1173 K and 400 K; the carrier is laminar throughout. With the baseline diffusivity, the radial diffusion time R^2/D_i is 58 ms in steel and 63 ms in silica. Placing the conductance of eq. (4) in series with diffusion over the tube radius, as in eq. (5) with d_f replaced by R and $T_c = T_w$, defines a wall Damköhler number

$$\begin{aligned} \mathrm{Da}_i &= \frac{G_i \chi_i R}{\rho D_i} \\ &= C_e s_i \frac{2\alpha_i}{2-\alpha_i} \sqrt{\frac{RT_w}{2\pi M_i}} \frac{R}{D_i}, \end{aligned} \tag{14}$$

where R is the gas constant under the square root and the tube radius in the last factor. Because $\chi_i/\rho = RT_w/M_i$, it does not depend on pressure. For PbI_2 and BiI_3 between 400 and 1173 K, the baseline coefficients give $\mathrm{Da}_i \approx 1.4 \times 10^3$–$2.9 \times 10^3$, and the tested accommodation coefficients $\alpha_i = 0.1$ and 0.01 give about 76–150 and 7–15. This tube-scale estimate places the interfacial resistance well below the radial diffusion resistance for $\alpha_i = 1$ and 0.1, and about an order of magnitude below it for $\alpha_i = 0.01$. It is a scaling estimate, not a substitute for the resolved near-wall gradients of the calculations.

Two effects that the model omits can be bounded with the same inputs, the digitised wall profiles and the computed absolute pressure of 101.33 kPa. The first is buoyancy. Neither experimental paper states the column orientation. For developed laminar flow with a linearly varying wall temperature, the wall-to-bulk temperature difference is $(11/192)\, U d^2 |\mathrm{d}T_w/\mathrm{d}x|/a$, with a the thermal diffusivity of helium; it stays below 2.6 K. A Grashof number based on the axial temperature change over one diameter, $\mathrm{Gr} = g d^4 |\mathrm{d}T_w/\mathrm{d}x|/(T\nu^2)$, stays below 2.2, and below 1.4 where the wall temperature lies between 400 and 750 K. Because Re is also of order one, the Richardson number $\mathrm{Gr}/\mathrm{Re}^2$ reaches 0.15 (pure PbI_2) to 0.78 (silica) in that range, and about 1.2 in the steepest cold section of the silica profile. Compositional buoyancy at the largest computed trace mole fraction, 9.2×10^{-5} evaluated with the molar mass of BiI_3, gives $\mathrm{Gr}/\mathrm{Re}^2 \leq 0.32$. Buoyancy-driven secondary flow is therefore weak, but in a horizontal column it cannot be excluded, and the frozen carrier does not represent it. The second effect is thermal (Soret) diffusion. Taking a thermal diffusion factor $\alpha_T = 1$ as a generous bound, the Soret drift velocity $D_i \alpha_T |\mathrm{d}T_w/\mathrm{d}x|/T$ is at most 2.2% of the carrier velocity. The radial Soret flux, relative to a Fickian flux $x_i D_i/R$ towards the wall, is at most $\alpha_T \Delta T_r/T \leq 0.7\%$, with ΔT_r the wall-to-bulk difference above. Both are small compared with the factor-of-two diffusivity variations examined in section 7.

4.2 Discretisation and coupling

The transient species equations use implicit Euler time integration and corrected Gauss linear diffusion. The production cases (the 15 baselines and four silica temperature-coordinate cases) use the bounded OpenFOAM scheme `Gauss limitedLinear 1` [16, 17] for advection on 41,472 cells. It blends linear and upwind interpolation through a TVD-type flux limiter, which reverts to upwinding near extrema and steep gradients. The 64 follow-up cases and the time-step comparisons keep the first-order upwind advection of the original campaign on 18,432 cells, and the follow-up sensitivities are always reported as differences from their own upwind baselines on that mesh. The scheme and mesh study adds the van Leer limiter [18] (`Gauss vanLeer`) as a second bounded scheme. All nine transported gases use the same advection scheme within a case. The baseline time step is 2 ms. Wall exchange is assembled implicitly with nonnegative source and sink coefficients and shared inventory bounds. Configured species correctors and active-bound guards reconcile the final gas solution with the realised wall transfer. Gas re-speciation then redistributes species under the local element constraints. The sequence is first order in time; a conservative equilibrium projection alone does not remove splitting or transport error.

The baseline configuration requests three species correctors with a 10^{-16} tolerance and up to eight bound guards with a 10^{-13} guard tolerance. These are stopping settings. They do not guarantee that every outer wall-regime iteration reaches the requested tolerance, so the final transport defect, bound status and material ledger are examined separately (section 5.1). Short corrector replays and three full runs with eight correctors test the effect of this stopping criterion (section 5.1).

Every gas has the baseline diffusivity $D_i = 10^{-4}\,\mathrm{m^2\,s^{-1}}$. Three transport alternatives halve or double this value for all gases, or replace only the PbI_2 coefficient with a temperature- and pressure-dependent PbI_2–He estimate. The collision parameters of that estimate are themselves estimates, not independently measured binary transport data.

Table 2: Completed follow-up matrix: number of cases and the variation applied in each study. The baseline cases serve as additional references. Each numerical comparison changes one resolution setting at a time; flow-stop cases continue a 60 s baseline to 80 s.

Study	Cases	Variation
Mesh and time step	12	Three grids; 2 and 1 ms
Gas diffusion	12	$D/2$, $2D$, PbI_2–He correlation
Iodide accommodation	8	$\alpha = 0.1$, 0.01
Source position	8	$\Delta x_s = \pm 1.5$ cm
Q2 metal loading	12	$\beta = 0.002$, 0.003162, 0.005623
Flow reference	4	273.15 versus 298.15 K
Silica temperature coordinates	4	$\Delta x_T = -4.10, -4.65$ cm
Flow-stop relaxation	4	60–80 s; fixed temperature

Accommodation variations set $\alpha = 0.1$ or 0.01 for the two reversible iodide pairs and keep the baseline metal-channel coefficients.

4.3 Completed case matrix

The 15 baseline cases are Q1 for each available configuration, plus Q2 for the two LBE columns at $\beta \in \{0.001, 0.01, 0.1\}$ with two thermodynamic branches. The second branch adds a constant 29 kJ/mol to the standard Gibbs energy of BiI(g). It is a sensitivity surrogate, not a second experimentally qualified dataset. All completed calculations use the same solver executable, identified by its SHA-256 in the saved reports.

Table 2 lists the 64 follow-up cases. The studies cover the following cases:

- mesh and time-step checks: pure PbI_2 and three steel Q2 cases (baseline thermodynamics at 0.1% and 1% metal saturation, and shifted BiI thermodynamics at 1%);
- physical transport, accommodation, source-position and flow-reference variations: pure PbI_2, the two steel Q2 cases with baseline thermodynamics, and silica Q2 at 0.1%;
- extra source-loading cases: both columns and both thermodynamic branches;
- temperature-coordinate variants: silica Q1 and silica Q2 at 0.1%.

This is a structured set of conditional comparisons, not a factorial uncertainty study.

The source-position alternatives move the release region by $\pm 1.5\,\mathrm{cm}$ without adding a boat obstruction. The flow-stop cases restart from the saved 60 s fields, remove injection, set the carrier flow to zero and impose zero normal species flux at both ends. The original nonuniform temperature profile stays fixed. These cases therefore measure how residual gas redistributes during a 20 s flow stop at fixed temperature; they are neither a clean-helium purge nor a full cooling calculation.

Four further sets of runs complete the evidence, all with the same solver executable. (i) The 19 production cases repeat the 15 baselines and the four silica temperature-coordinate cases on 41,472 cells with the limited-linear scheme; all other inputs are unchanged. (ii) A scheme and mesh study takes two cases, pure PbI_2 and the shifted-BiI steel case at 1% metal saturation, which had the largest upwind fine/medium change. It runs them with upwind, limited-linear and van Leer advection on 8,192 to 41,472 cells, and the steel case with upwind and limited-linear also on 93,312 cells. This adds 12 runs to the existing upwind levels and the production runs. (iii) Three original 18,432-cell upwind cases are repeated over the full 60 s with eight instead of three species correctors. (iv) Ten runs simulate five single-species PbI_2 experiments of Lobresco et al. [6] on two meshes (section 6.3).

5 Numerical verification and resolution limits

This section reports material conservation and solver diagnostics, then the separate time-step and mesh comparisons, and finally two effects that limit how precisely local profile features are resolved.

5.1 Completion, material balance and solver diagnostics

All 15 runnable baseline cases, all 64 follow-up cases, the 19 production cases and the three full-run corrector cases completed and passed their configured numerical acceptance checks; the scheme and mesh runs completed with the

same ledgers. This PASS classification means only that the computation completed and passed those checks; it is not an experimental-validation classification. Element conservation is examined in two ways: through the solver ledger and through an independently formed material residue. For element k the residue is

$$R_k = N_k^{\mathrm{gas}} + N_k^{\mathrm{wall}} + N_k^{\mathrm{sample}} + N_k^{\mathrm{net\,out}} + N_k^{\mathrm{removed}} - N_k^{\mathrm{initial}} - N_k^{\mathrm{released}}, \tag{15}$$

where N_k are molar atom inventories and outward transport is signed. The ledger also records the linear-solve defect and any clamping correction separately. These terms explain bookkeeping closure, but they are not inserted into R_k when the raw material balance is assessed.

All residues below are divided by the supplied material and taken as the largest value over recorded steps and represented elements. The largest raw residue is 1.41×10^{-12} for the 19 production cases (41,472 cells, limited-linear advection) and 8.85×10^{-13} for the 60 source-driven follow-up cases; the 18,432-cell upwind baselines had 5.97×10^{-13}. For the follow-up cases, the largest accounted closure is 9.60×10^{-16} and the largest recorded solver defect is 8.64×10^{-13}; for the production cases they are 1.12×10^{-15} and 1.45×10^{-12}. The four flow-stop cases keep their restart ledger and are excluded from this summary of newly accumulated source material. These results support conservation of the executed discrete model. They do not quantify physical accuracy or discretisation error.

The baseline logs report no gas-equilibrium solver failures and a maximum equilibrium element residue of 6.43×10^{-16}. Input-validity and solver-success counters are distinct. At startup, sign flags caused equilibrium to be skipped in up to 1,170 cells on one step, and the maximum negative iodine diagnostic was 1.14×10^{-124} of supplied iodine. Matched five-step startup replays identified extremely small leading-tail inventories at these flags; no flagged cells remained at the final step. The calculations should therefore not be described as free of negative trace values, even though the reported amount is immaterial to the macroscopic balance. The sum of the individual species mole-fraction maxima was at most 9.19×10^{-5}. Because these maxima need not coincide, this is an upper bound on the reactive trace loading, not a measured mixture mole fraction at one location.

Some paired wall-coupling iterations did not meet the requested 10^{-16} outer criterion. This is reported separately from equilibrium failure, the final transport defect and the inventory-bound checks. To test its effect, matched restarts of four representative cases at 59 s compared three and eight correctors, and outer tolerances of 10^{-16} and 10^{-14}, over 20 time steps (40 ms). The largest species-field change, scaled by its control maximum, was below 3.6×10^{-8}, and the changes in new wall increments were negligible. Three cases were then repeated over the full 60 s with eight instead of three correctors (18,432 cells, upwind). The iodine profiles of pure PbI_2 and of the steel Q2 case at 0.1% saturation change by $1.09 \times 10^{-13}\%$ and $4.89 \times 10^{-13}\%$, that is, at round-off; the steel case at 1% saturation changes by 0.021%. With eight correctors the largest outer initial residual is still 2.4×10^{-15}–4.4×10^{-15}, and nearly every step ends at the corrector limit, as with three. The requested 10^{-16} lies below the attainable residual floor, and the iteration error of the reported profiles is negligible next to the time-step and spatial errors below. No manufactured-solution test is presented. Section 6.3 compares the model with published measurements and published model results; it is not a code-to-code verification with identical inputs.

5.2 Separate time-step and mesh comparisons

For profiles on the common 1 cm bins, let a_b be the deposited iodine atoms in bin b for a comparison case and a_b^0 the corresponding baseline inventory. We report the profile change E_{L1} and the total change ΔN_I:

$$E_{L1} = 100 \frac{\sum_b |a_b - a_b^0|}{\sum_b a_b^0}, \qquad \Delta N_I = 100 \frac{\sum_b a_b - \sum_b a_b^0}{\sum_b a_b^0}. \tag{16}$$

Both measures retain changes in amount. The figures also show normalised shape curves; these must not be confused with the unnormalised profile change. Source, physical coefficients, equilibrium engine and executable are fixed within each numerical comparison.

Table 3 separates time-step from mesh and scheme effects. For successive mesh levels the reference a_b^0 is the finer level. Reducing the time step from 2 to 1 ms on 18,432 cells changes the iodine profile by 0.0043–0.3228%. With the upwind scheme of the original campaign, the 18,432 to 41,472-cell differences of the four follow-up cases are 2.06–10.58%, and the shifted-BiI steel case changes by essentially the same amount on the first two refinements (10.47% and 10.58%). Total deposited iodine changes by less than 0.014% from 18,432 to 41,472 cells, and the centroid moves by at most 0.0652 cm. Integral quantities are thus stable while local profile shapes still differ appreciably.

The scheme and mesh study (table 3 and fig. 3) addresses the shifted-BiI steel case directly. With the limited-linear scheme its iodine profile changes by 15.35, 8.95 and 1.98% over the three refinements to 93,312 cells; with upwind, by 10.47, 10.58 and 2.13%. Pure PbI_2 changes by 5.70 and 2.15% with limited-linear, against 6.57 and 5.06% with

Table 3: Iodine profile changes E_{L1} (percent) between successive mesh levels at 2 ms, relative to the finer level, for three advection schemes, and the effect of halving the time step on 18,432 cells with upwind advection. All steel cases are Q2; the shifted-BiI case uses 1% metal saturation. The upwind levels of the two lower steel cases are the follow-up cases of table 2. A dash marks a combination that was not run.

Case	Scheme	8,192→18,432	18,432→41,472	41,472→93,312	2→1 ms
Pure PbI_2	upwind	6.57	5.06	–	0.0043
	limitedLinear	5.70	2.15	–	–
	van Leer	4.94	2.23	–	–
Steel, shifted BiI	upwind	10.47	10.58	2.13	0.1536
	limitedLinear	15.35	8.95	1.98	–
	van Leer	16.83	9.17	–	–
Steel Q2, 0.1%	upwind	9.02	5.26	–	0.2326
Steel Q2, 1%	upwind	9.72	2.06	–	0.3228

upwind. The van Leer scheme gives successive changes close to those of limited-linear and differs from it by at most 2.8% (steel) and 1.8% (pure PbI_2) at equal mesh, and by 1.16% and 0.27% on 41,472 cells. Between 41,472 and 93,312 cells, the 3 cm smoothed profile of the steel case changes by 0.67% and its total deposit by 0.015%; over all four levels its centroid stays within 0.6 mm and the wall temperature at the PbI_2 centroid within 0.9 K.

Three-grid scalar checks follow the distinction between refinement trends and an established asymptotic range [19–21]. For the earlier upwind follow-up triplets, 17 of the 34 nontrivial or zero-valued combinations of species, total, centroid and width are oscillatory, four have nondecreasing differences, one is unresolved at roundoff and 12 admit only conditional Richardson/GCI estimates. For the limited-linear steel sequence, the iodine total, centroid and width oscillate on both triplets, and the successive profile changes give apparent orders of 1.3 and 3.7. These orders are inconsistent, so no formal grid convergence index is claimed. We quote the measured 41,472 to 93,312-cell change of 1.98% as the spatial uncertainty indicator of the production profiles. If the last change were extrapolated at the last apparent order, the 41,472-cell profile would lie about 2.5% from the mesh limit; assuming second order, about 3.6%. These are indicative estimates for one case, not confirmed uncertainty bounds, and they do not establish mesh independence of local profile shapes in other cases.

5.3 Binned maxima and numerical diffusion

A deposition maximum picked on a 1 cm grid is fragile when adjacent bins have similar heights. In pure PbI_2 the 18,432-cell upwind maximum is at 37 cm and 350.7 °C, while the 36 cm bin is only about 0.05% lower. On the 41,472-cell upwind mesh the maximum moves to 36 cm and 366.5 °C, although the centroid shifts by only -0.0585 cm. With the production scheme (41,472 cells, limited-linear) the binned maximum stays at 36 cm and 366.4–366.5 °C on all three levels, and on 41,472 cells the 37 cm bin is 23% lower, yet the maximum of the 3 cm smoothed profile is at 37 cm and 350.5 °C, and the wall temperature at the deposit centroid (37.11 cm) is 348.7 °C. In the limited-linear steel sequence the PbI_2 maximum alternates between two nearly equal bins, 50 and 51 cm (306.3, 288.1, 306.7 and 288.1 °C on the four levels), while the 3 cm smoothed maximum stays at 51 cm and 288.0–288.1 °C and the centroid temperature at 290.6–291.5 °C. In the upwind steel Q2 0.1% case, the dominant BiI_3 bin moves from 57 to 58 cm, and its reported temperature changes from about 166.4 to 138.8 °C. A one-bin switch can therefore change the peak temperature by nearly 28 K without a comparable displacement of the profile as a whole. Peak temperature alone is an insufficient convergence metric; table 4 therefore also reports the smoothed and centroid values.

First-order upwind advection, used in the original campaign and the follow-up cases, smooths the profiles. The local one-dimensional estimate $D_{\mathrm{num}} \simeq |u_x|\Delta x/2$ is comparable to or greater than the prescribed $10^{-4}\,\mathrm{m^2\,s^{-1}}$ diffusivity in parts of the carriers. This is a Taylor-series screening estimate, not a measured truncation error for the radial, variable-density coupled problem, but the scheme study confirms its consequence. At equal mesh, upwind differs from limited-linear by 20.8, 14.1, 11.3 and 7.7% for the steel case on the four levels and by 9.0–13.2% for pure PbI_2, much more than the two bounded schemes differ from each other. Upwind numerical diffusion is therefore the dominant spatial error of the original campaign, and it is why halving the time step could not establish convergence of profile widths. Across the 19 production cases, the production profiles differ from the 18,432-cell upwind profiles by 9.1–22.3%. The follow-up sensitivities of section 7 remain upwind differences on 18,432 cells, taken against baselines with the same numerical error.

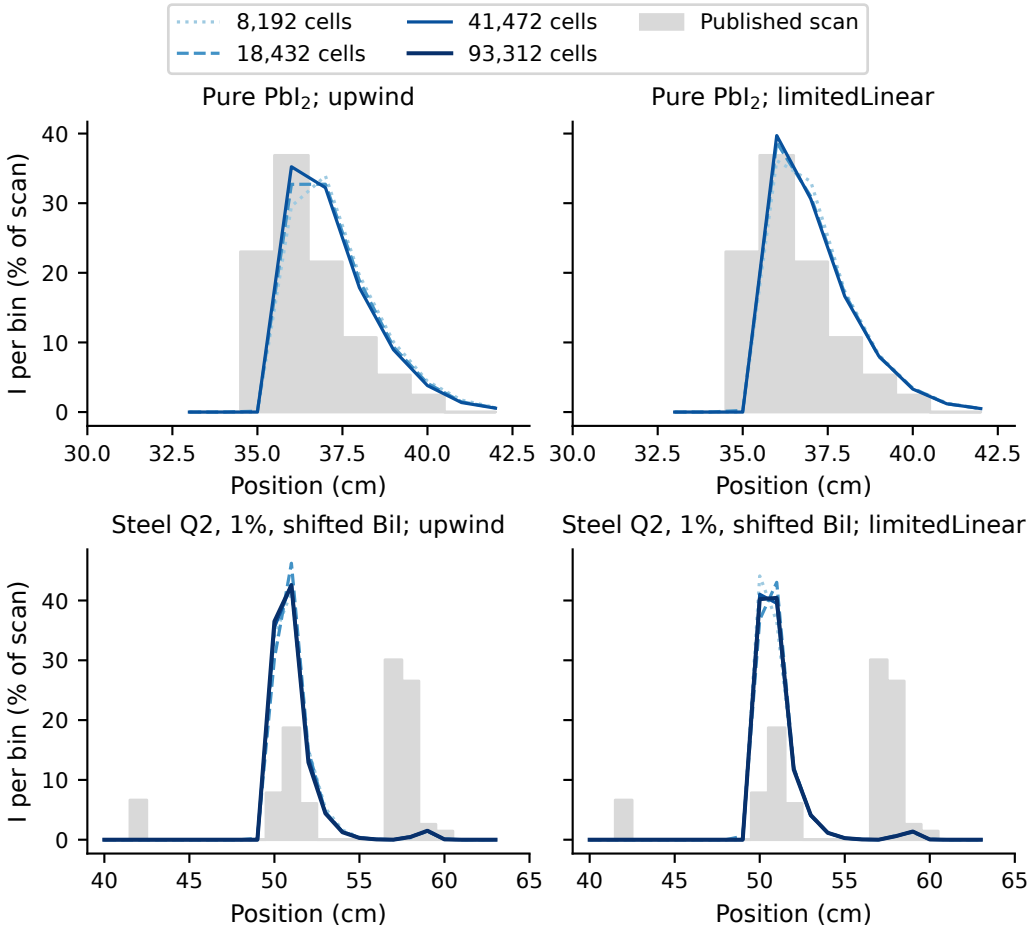


Figure 3: Mesh sequences of the iodine deposition profile for pure PbI_2 (top) and the shifted-BiI steel case at 1% metal saturation (bottom), with first-order upwind (left) and limited-linear (right) advection. Each curve is normalised over the common scanned interval so that shapes can be compared; the unnormalised differences are given in table 3. Grey histograms are the reconstructed experimental iodine scans [1] and include any KI contribution.

6 Conditional comparison with deposition measurements

This section compares the calculations, mainly the original Q1 cases, with the reconstructed deposition scans. The comparisons are conditional on the reconstructed inputs and assumptions of sections 2 and 3; they are not independent validation.

6.1 Comparison measures

The primary comparison uses reconstructed relative iodine yields per 1 cm bin. Let e_b be the experimental percentage and let

$$m_b = (100 - y_{\mathrm{KI}}) \frac{a_b}{\sum_{\text{full column}} a_b} \tag{17}$$

be the modelled percentage, where the stated excluded KI yield is $y_{\mathrm{KI}} = 0$, 6.6 or 7.0. Over the common scanned bins $\mathcal{B}$, we report the raw discrepancy

$$L_{\mathrm{raw}} = \sum_{b \in \mathcal{B}} |m_b - e_b| \quad \text{(percentage points)}. \tag{18}$$

This measure keeps the experimental digitisation sum and the model's explicit KI exclusion. It is neither an absolute deposited-mass error nor a percentage relative to an experimental uncertainty.

Table 4: Q1 iodide maxima on the production mesh (41,472 cells, limited-linear advection), compared with the published peaks. Subscript bin: 1 cm binned maximum; subscript 3: maximum of the 3 cm smoothed profile; T_c: wall temperature at the deposit centroid. Positions are in cm and temperatures in °C. Experimental temperature ranges are those printed by Liu et al. [1]; they are not statistical confidence intervals for model error.

Case	Species	x_{exp}	x_{bin}	x_3	T_{exp}	T_{bin}	T_3	T_c
Pure PbI_2, steel	PbI_2	36	36	37	368±18	366.5	350.5	348.7
LBE, steel	PbI_2	51	51	52	288±18	288.1	269.2	274.5
LBE, steel	BiI_3	57	57	58	154±14	166.9	138.8	158.1
LBE, silica	PbI_2	51	56	56	306±24	287.8	287.8	288.8
LBE, silica	BiI_3	59	63	64	127±39	152.8	119.8	147.1

A secondary shape diagnostic normalises both profiles over $\mathcal{B}$, $\widehat{m}_b = m_b / \sum_{\mathcal{B}} m_b$ and $\widehat{e}_b = e_b / \sum_{\mathcal{B}} e_b$, and computes the overlap

$$O = 100 \sum_{b \in \mathcal{B}} \min(\widehat{m}_b, \widehat{e}_b). \tag{19}$$

The overlap O is conditional on the common scanned range and discards differences in capture and total yield. The experimental normalisation still contains KI. O is a descriptive shape measure, not a validation probability or a fitted goodness-of-fit statistic.

6.2 Q1 peaks and complete profiles

Table 4 compares the five available Q1 iodide maxima with the printed peak temperatures. Besides the 1 cm binned maximum it lists the maximum of the 3 cm smoothed profile (x_3, T_3) and the wall temperature at the deposit centroid (T_c). All calculated binned peak temperatures lie within the stated temperature ranges. For pure PbI_2 the binned maximum (36 cm, 366.5 °C) matches the published 36 cm and 368 °C, but the smoothed maximum and the centroid temperature (350.5 and 348.7 °C) lie at the low-temperature edge (350 °C) of the printed range. For steel PbI_2 the binned value is 288.1 °C, whereas the smoothed and centroid temperatures, 269.2 and 274.5 °C, lie at the lower limit of the printed 288±18 °C. The steel BiI_3 smoothed maximum (138.8 °C) falls just below its range, while the centroid temperature (158.1 °C) lies inside it. The agreement of a binned peak temperature therefore depends on which of two nearly equal bins is the larger (section 5). Steel LBE places both iodide maxima at the measured positions, whereas silica LBE places the PbI_2 and BiI_3 maxima 5 and 4 cm downstream, respectively. Agreement with a printed temperature is therefore not sufficient to establish the correct spatial location.

Figure 4 shows the complete profiles. The raw discrepancies L_{raw} are 46.33, 52.97 and 183.84 percentage points for pure PbI_2, steel LBE and silica LBE (53.90, 53.45 and 181.84 on the 18,432-cell upwind mesh), and the corresponding normalised overlaps are 76.97%, 72.13% and 6.13% (72.93%, 71.88% and 7.20%). The silica case therefore has a large profile discrepancy even though both printed temperature ranges contain its calculated maxima.

The Q1 steel BiI_3 share of deposited iodide iodine is 65.34%, close to the measured iodide-only share of $61.3/(61.3 + 32.1) = 65.63\%$. For silica the corresponding values are 28.68% and $27/(27 + 66) = 29.03\%$. These agreements are substantially conditioned by the source masses, which were inferred from the same deposits. They cannot serve as independent validation of gas speciation or LBE release. Similarly, the source-driven calculations capture 91.28–98.20% of the supplied iodine on the wall by 60 s. This does not establish the experimental three-hour capture, because the remaining inventory, the release history and post-release cooling are different questions.

6.3 Independent single-species comparison: Lobresco et al.

The Liu et al. comparisons above are conditioned on deposit-derived sources. A cleaner test of the transport and wall-exchange model is offered by the five PbI_2 thermosublimatography experiments of Lobresco et al. [6], which vary the helium flow (5, 106 and 540 mL/min) and the duration (6, 60 and 300 s) with 1.37–2.23 mg samples in a 4.8 mm steel column. That study also reports deposition temperatures from two of its own CFD models: a temperature-based model calibrated against one reference experiment, and a vapour-pressure-based Hertz–Knudsen–Schrage model with a near-wall interaction layer and an effective rate multiplier C_e. We ran the five cases without adjusting any parameter to these data and compare all three models with the measurements by the paper's own rule: the deposition temperature is the wall temperature at the first position, from upstream, where the deposit exceeds 1% of its peak, linearly interpolated between 1 cm points.

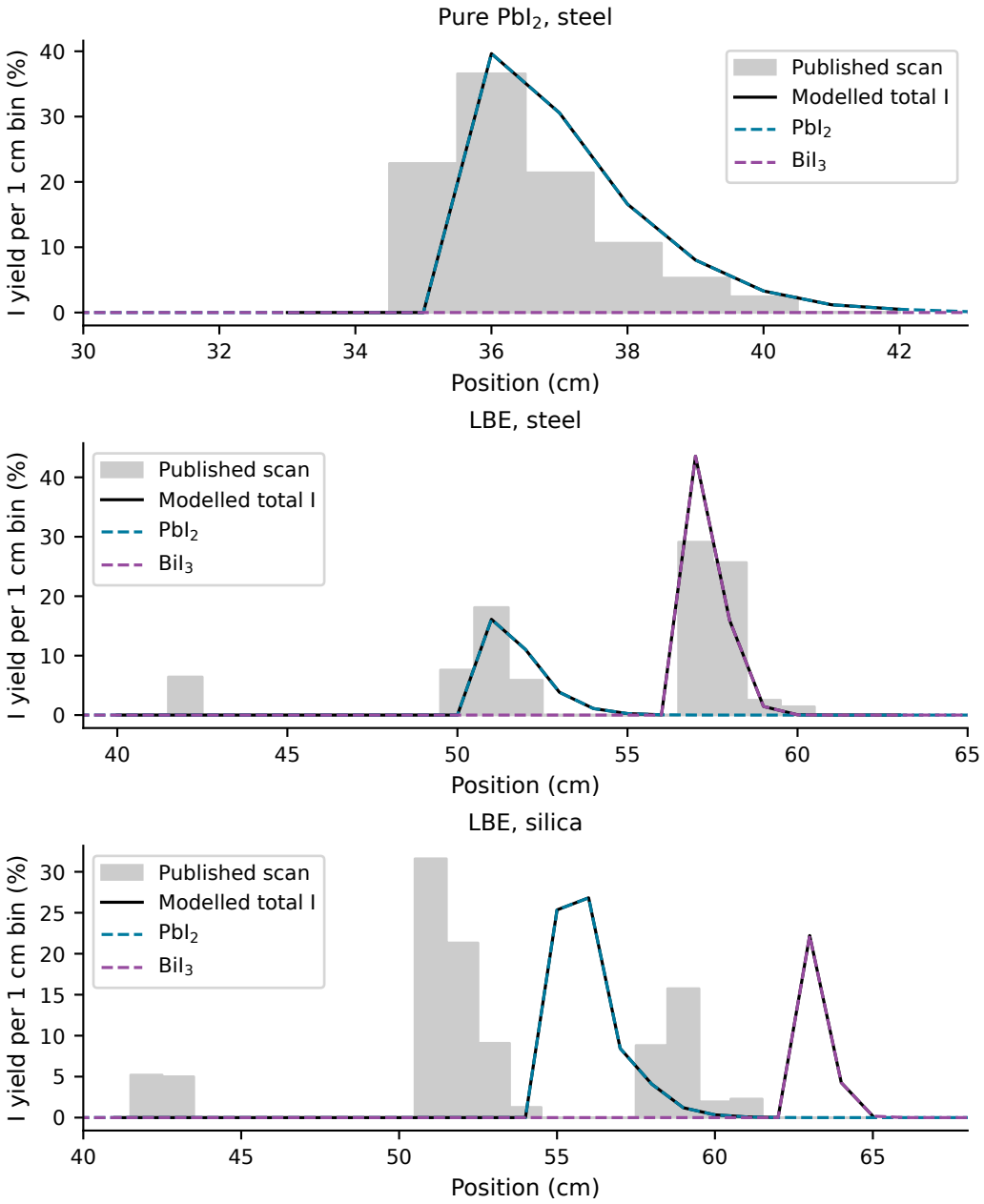


Figure 4: Production Q1 iodine profiles (41,472 cells, limited-linear advection) compared with the digitised published scans (grey histograms) [1]. Solid black lines are total modelled wall iodine; dashed lines show its two iodide contributions. Model totals are normalised to 100%, 93.4% and 93.0% for pure PbI_2, steel LBE and silica LBE, respectively, to expose the omitted KI yield; experimental bars keep their digitised values.

Inputs. Table 1 of the paper gives the flows, durations and sample amounts; its flows are standard volumetric rates at 273.15 K and 101,325 Pa. The wall temperature of each flow rate, the measured deposition profiles, the sample position and both published model curves were digitised from its Figs. 13–14 (CC BY 4.0). Applying the 1% rule to the digitised profiles reproduces the published experimental deposition temperatures within 1.0 K. A wall temperature table distributed with the T-Flows test case of that study differs from these published curves: its maximum of 1210 K lies at 0.04 m, whereas the figures peak at 1060–1070 K at 0.13–0.15 m, and between 0.30 and 0.60 m it differs from them by up to 68–92 K. We use the published curves. Everything else is our model as described in sections 3 and 4, reduced to one species: an expanding laminar helium carrier computed with the wall temperature, HKS exchange with $C_e = \alpha = 1$ in series with the half-cell diffusion resistance, the PbI_2–He diffusivity correlation of the same paper, and

Table 5: Deposition-onset temperatures for the five PbI_2 experiments of Lobresco et al. [6]: flow $\dot{V}$ (mL/min, standard), duration t (s), measured onset T_{exp} (K, their Table 2), and the onset of this work and of their temperature-based and vapour-pressure-based models minus T_{exp} (K; the latter two from their Table 2). Δx_{pk} is our deposit peak minus the measured peak (cm), and L_1/n_0 the summed absolute difference of our deposit and the digitised measured profile divided by the sample amount. This work: 22,896 cells, deposit at the end of the experiment; for T5_60 the main deposit downstream of the sample.

Case	$\dot{V}$	t	T_{exp}	This work	Temp.-based	Vap.-press.	Δx_{pk}	L_1/n_0
T106_6	106	6	814	+26	−39	+27	−4	0.89
T106_60	106	60	723	+19	−10	+37	−2	0.66
T106_300	106	300	667	+16	−1	+34	−2	0.88
T5_60	5	60	847	−11	−60	−50	−1	0.37
T540_60	540	60	700	−14	−30	−15	−1	0.47
RMS				18	35	35		

our own phase-resolved PbI_2 vapour-pressure table. The sample is an evaporating inventory of the Table 1 amount, 1 cm long and 0.48 mm in radius at the digitised position; this geometry is an assumption, but the sample empties in less than 0.5 s, so release is not rate-limiting. The 0.69 m wedge has 22,896 cells at the axial and radial spacing of the production mesh and 10,176 cells at that of the 18,432-cell mesh, with limited-linear advection, a zero-total-flux inlet that lets vapour diffuse upstream, 2 ms time steps (0.5 ms at 540 mL/min) and the sample removed at the end of the experiment.

Results. Table 5 and fig. 5 summarise the comparison. Our onset temperatures differ from the measurements by +26, +19, +16, −11 and −14 K, an RMS deviation of 18 K, against 35 K for the calibrated temperature-based model and 35 K for the vapour-pressure-based model of the paper. The two meshes agree within 0.2 K. The experimental decrease of the onset temperature with duration at 106 mL/min is reproduced, with a warm offset of 16–26 K, and the onsets at 5 and 540 mL/min are 11–14 K too cold. Three of the five deviations exceed the stated ± 15 K experimental temperature uncertainty. The agreement in profile shape is only moderate: the computed peaks lie 1–4 cm upstream of the measured peaks, and the absolute profile difference is 0.4–0.9 of the sample amount.

Three qualifications apply. First, at 5 mL/min vapour diffuses upstream against the slow carrier and deposits near the cold inlet; applied from the inlet, the 1% rule then picks this small deposit, at 548 K. We apply the rule to the main deposit downstream of the sample. The paper's vapour-pressure model shows a similar upstream deposit, so its Table 2 value for this case evidently refers to the main deposit as well. Second, the experiments end by stopping the flow, withdrawing the sample and pulling the column out of the furnace; this transient is not simulated. Counting the vapour still in the tube at the end time where it is changes only T5_60 noticeably, to +3 K, because 4.0% of its inventory is still gaseous (RMS 17 K). Third, next to the sample the computed PbI_2 mole fraction reaches 0.33–0.36, far outside the dilute, passive-carrier assumption of the model; the boat itself is not represented. The comparison therefore tests the model in a regime it was not built for near the source, and it rests on digitised rather than tabulated wall temperatures. It is an independent comparison with published measurements and published models, not a validation.

7 Input sensitivity and interpretation

This section examines how source loading, BiI thermodynamic data, the silica temperature coordinate, and transport, kinetic and flow inputs change the computed deposits.

7.1 Source loading and thermodynamic information

The Q2 cases expose a sensitivity that is hidden when the iodide split is prescribed. Figure 6 shows the BiI_3 share of deposited iodide iodine,

$$f_{\mathrm{BiI_3}} = \frac{3N^{\mathrm{wall}}_{\mathrm{BiI_3}}}{2N^{\mathrm{wall}}_{\mathrm{PbI_2}} + 3N^{\mathrm{wall}}_{\mathrm{BiI_3}}}. \tag{20}$$

This share counts only the iodine in the two modelled deposited iodides. It differs from the BiI_3 fraction of all experimental iodine, which also contains KI. Additional metal-saturation cases fill selected points between the original 0.1% and 1% levels; line segments connect computed points and are not a fitted release law.

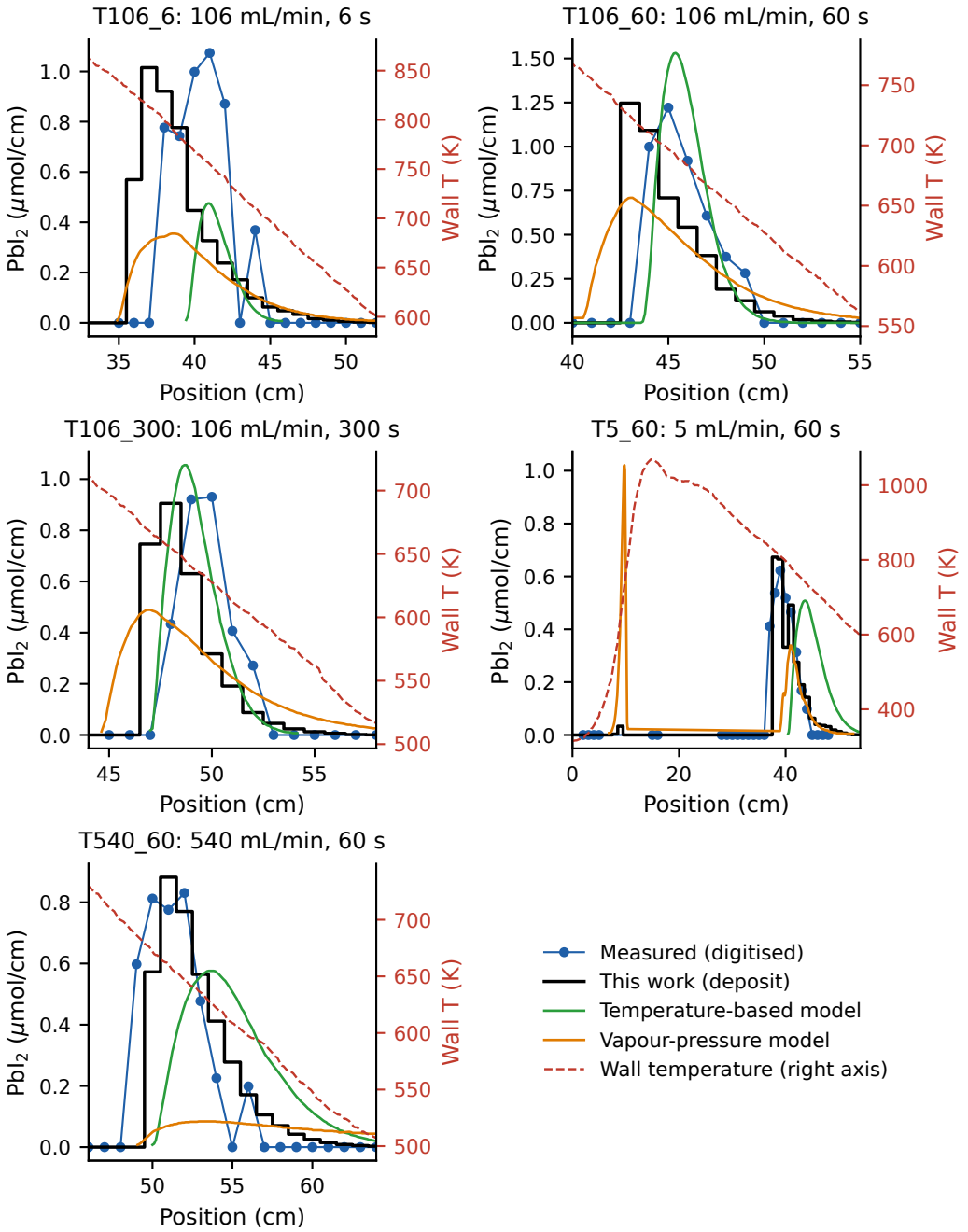


Figure 5: Deposited PbI_2 for the five experiments of Lobresco et al. [6]: measured profiles, their temperature-based and vapour-pressure-based model results (all digitised from their Figs. 13–14 and adapted under CC BY 4.0), and the deposit of this work at the end of each experiment (1 cm bins, 22,896 cells). Dashed red lines: digitised wall temperature used as input (right axes).

At 1% metal saturation on the production steel mesh, the baseline thermodynamics give $f_{BiI_3} = 94.58\%$ (94.57% with upwind advection). Raising the BiI(g) standard Gibbs energy by 29 kJ/mol gives 2.01% (2.00%) at the same numerical resolution. This large difference survives refinement. It therefore reflects conditional thermodynamic sensitivity, not numerical noise. The perturbation is applied consistently to the formation and reactive-wall constants. A constant Gibbs-energy shift scales the formation constant by $\exp[-29{,}000/(RT)]$, approximately 0.051 at 1173 K and 0.00093 at 500 K. It is a substantial surrogate perturbation, not an independently measured correction to BiI thermodynamics.

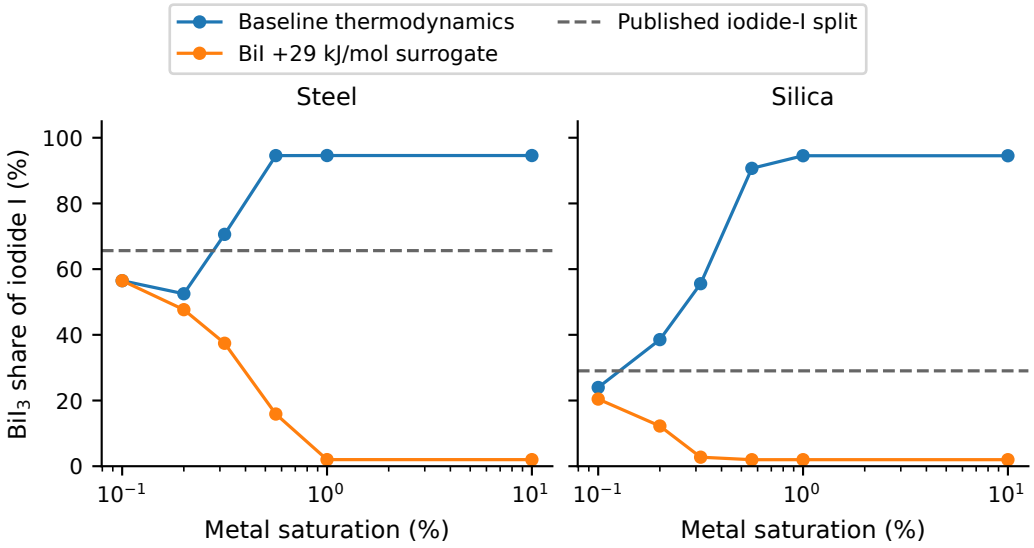


Figure 6: Conditional sensitivity of the BiI_3 share of deposited iodide iodine (eq. (20)) to source loading and BiI thermodynamics, for steel and silica LBE. Symbols denote completed Q2 cases; segments only connect those points. Dashed grey lines show the published BiI_3 share after excluding KI [1]. The +29 kJ/mol branch is a surrogate perturbation, not a second qualified thermodynamic dataset.

Source loading also matters. With baseline thermodynamics, the steel BiI_3 share at 0.2%, 0.316% and 0.562% metal saturation (follow-up cases, 18,432 cells, upwind) is 52.50%, 70.58% and 94.56%; the shifted branch gives 47.66%, 37.43% and 15.88%. The low-loading trend is not universally monotonic. The original steel 0.1% case with baseline thermodynamics gives 56.48% and an 80.66% scan overlap (56.47% and 81.10% on the 18,432-cell upwind mesh). This makes it a useful diagnostic comparison, but not an identified physical saturation. The published deposits alone cannot uniquely separate source loading, thermodynamic data and omitted chemistry. Choosing a branch because it best matches one scan would introduce calibration and would require independent validation.

7.2 Silica temperature-coordinate discrepancy

In the silica experiment, the published peak labels do not match the published temperature curve. The scan places the PbI_2 and BiI_3 maxima at 51 and 59 cm, labelled 306±24 and 127±39 °C. At those positions the digitised temperature curve gives approximately 369 and 235 °C, which differ from the labels by 63 and 108 K. On the same curve, 306 and 127 °C occur at 55.096 and 63.653 cm. The discrepancy exceeds the nominal raster temperature resolution and suggests a temperature-coordinate inconsistency in the supplied inputs. It does not identify which measurement, annotation or registration needs correction.

Two diagnostic mappings shift only the silica temperature curve upstream, by 4.096 or 4.653 cm, so that one of these curve crossings aligns with the corresponding labelled position. Equivalently, $T_\delta(x) = T_0(x + \delta)$ for $\delta > 0$, with $\Delta x_T = -\delta$. The iodine scan stays fixed. The shifts are derived from the published curve and labels, not optimised against the CFD profiles. All original unshifted results are retained.

Table 6 and fig. 7 show that the shifts move the Q1 maxima from 56/63 cm to 51/59 or 51/58 cm and raise the normalised overlap from 6.13% to 77.80% or 73.26%. The low-loading Q2 variants place the maxima at 51/59 or 51/58 cm, with overlaps of 77.88% and 75.54%. On the 18,432-cell upwind mesh the order of the two shifts was the opposite (75.29% and 81.00% for Q1; 79.16% and 83.33% for Q2), so the ranking of the two hypotheses by overlap is not robust to the advection scheme and should not be used to prefer one of them. A plausible coordinate change thus has a larger effect than the tested source-position or time-step changes. This does not confirm either shift as the experimental truth. The two shifts also give different peak temperatures, and no uniquely established mapping reconciles both original temperature labels at once.

The shifted low-loading Q2 cases keep a BiI_3 share near 23.97%, about five percentage points below the published iodide-only split. Correct peak positions therefore do not ensure the correct partition. Resolving the discrepancy requires the original thermocouple profile and its coordinate registration, not selection of the CFD variant with the highest overlap.

Table 6: Silica temperature-coordinate hypotheses: computed peak positions and agreement with the fixed experimental scan for the original and two shifted temperature curves. Shifts and peak positions are in cm; raw L1 (L_{raw}) is in percentage points and overlap in percent. Q2 uses baseline thermodynamics and 0.1% metal saturation.

Source	Δx_T	$x_{\mathrm{PbI_2}}$	$x_{\mathrm{BiI_3}}$	Raw L1	Overlap
Q1	0.00	56	63	183.84	6.13
Q1	-4.10	51	59	40.17	77.80
Q1	-4.65	51	58	51.44	73.26
Q2	0.00	55	63	183.84	6.13
Q2	-4.10	51	59	37.40	77.88
Q2	-4.65	51	58	47.21	75.54

Table 7: Range of the baseline-relative iodine profile change E_{L1} (percent) within each completed sensitivity category. The flow-stop row compares 80 s inventories with their 60 s restart state; the other rows compare 60 s inventories. The silica temperature-coordinate row uses the production cases (41,472 cells, limited-linear advection); all other rows are upwind follow-up cases on 18,432 cells compared with their own upwind baselines. Case counts and variations are listed in table 2.

Assumption	Minimum	Maximum
Gas diffusion	0.15	14.20
Iodide accommodation	0.25	4.21
Source position	0.08	0.19
Q2 metal loading	0.00	112.14
Flow reference	5.09	10.93
Silica temperature coordinates	187.10	193.96
Flow-stop relaxation	1.10	4.63

7.3 Transport, kinetics, flow reference and flow stop

Table 7 reports, for each tested category, the range of iodine profile changes E_{L1} (eq. (16)) from the designated baseline. These are responses to finite hypothetical changes, not statistical confidence limits. The source-loading row uses its 1% reference case. The temperature-coordinate row measures changes from the original silica input and can approach 200% when the profile supports move apart. These baseline-relative changes must not be confused with the raw discrepancies against the experimental scan.

Diffusion alternatives change profiles by up to 14.20%. A 273.15 K flow reference raises the inferred helium mass flow by 9.15% relative to 298.15 K and changes profiles by up to 10.93%. Iodide accommodation changes give up to 4.21%, whereas moving the prescribed source region by ± 1.5 cm gives at most 0.19%. This small source-position response concerns the selected empty-tube source model; it cannot rule out an effect of a real boat on the flow or thermal fields. Figure 8 shows the maxima for these categories. They are differences between upwind cases on the same 18,432-cell mesh, which itself differs from the production scheme by 12–14% for the two cases of the scheme study (section 5); how much of that error cancels in the differences has not been quantified. Where numerical and input changes have comparable magnitude, the present results do not rank their physical importance precisely. The tube-scale Damköhler numbers of section 4 and the small accommodation response, at most 0.41% for $\alpha = 0.1$ and 4.21% for $\alpha = 0.01$, against 3.83–14.20% for halving or doubling D, indicate that radial diffusion, not interfacial kinetics, limits the wall exchange in these columns; the wall temperature acts mainly through the equilibrium pressure.

The four fixed-temperature flow-stop cases change the deposited iodine profiles by 1.10–4.63% over 20 s, with centroid shifts of at most 0.0422 cm. Residual gas can therefore alter wall inventories after release stops. Because these runs neither cool the column nor flush it with clean helium (section 4), they do not predict the actual end-of-experiment redistribution after the furnaces are turned off. Likewise, equilibrium speciation and the irreversible BiI wall channel remain unqualified kinetic approximations.

8 Conclusions

We have applied a conservative finite-volume treatment of nine Pb–Bi–I gases and four shared condensed wall inventories in 15 baseline and 64 follow-up cases, 19 production re-runs and a scheme, mesh and iteration study. The columns are treated as laminar helium flows ($\mathrm{Re} \approx 0.7$–3) in tubes with a steep axial wall-temperature gradient, where axial advection, radial diffusion to the wall and a reversible interfacial exchange law together set the deposition profile. The

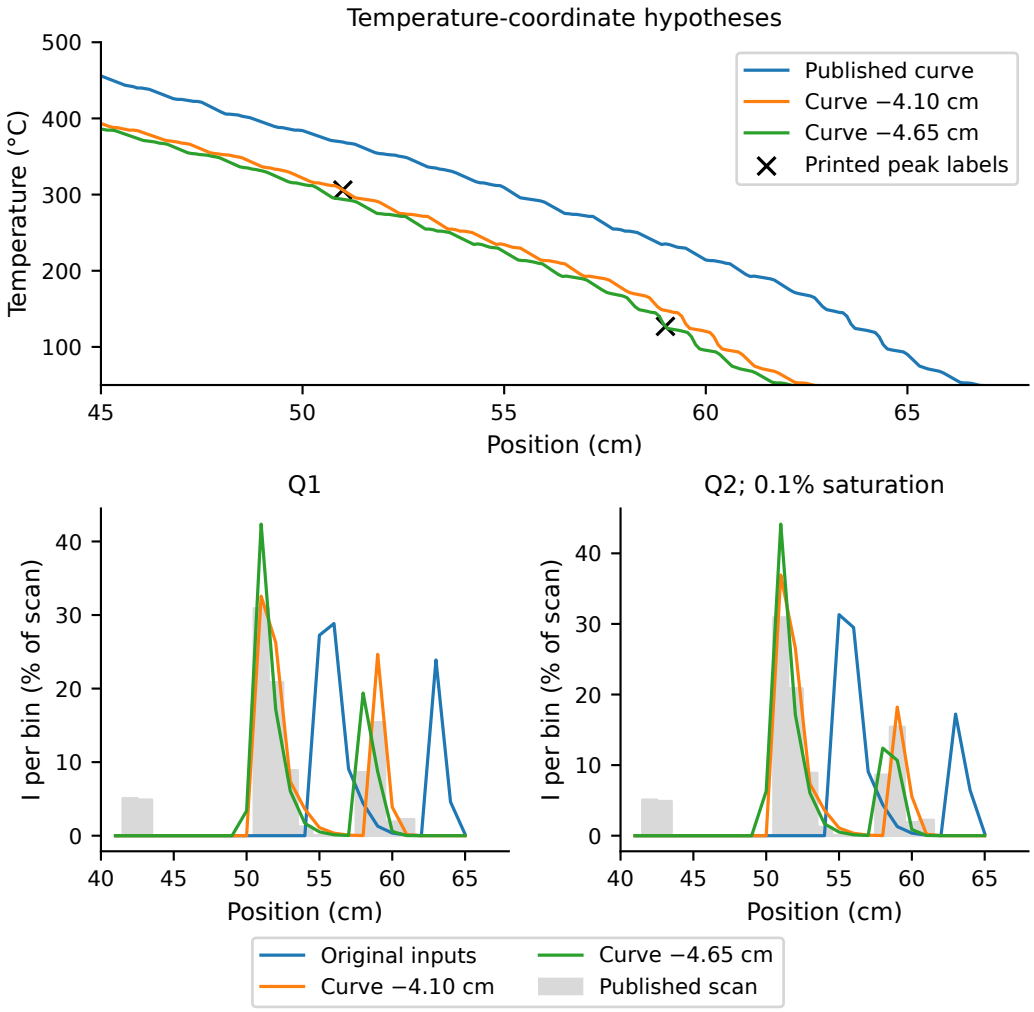


Figure 7: Input-derived silica mappings and their consequences. Top: the original reconstructed temperature curve, the two shifts and the printed peak labels. Bottom: common-scan-normalised iodine profiles for Q1 and low-loading Q2. The improved agreement is conditional on an unconfirmed coordinate hypothesis. Published data are adapted from Liu et al. [1].

reported calculations use a specialised ideal-gas equilibrium kernel, a prescribed helium carrier and reversible iodide wall exchange with half-cell diffusion resistance. Material bookkeeping is tight: the worst raw element residue is 8.85×10^{-13} of supplied material in the follow-up cases and 1.41×10^{-12} in the production cases. Conservation is a numerical property of this model; it is not evidence that all physical pathways are represented. Within the model, radial diffusion rather than interfacial kinetics limits the wall exchange: the wall Damköhler number is large, and the profiles respond far more to the diffusivity than to the accommodation coefficient.

Integral results are much less sensitive to resolution than local profiles. Halving the time step changes iodine profiles by at most 0.323%, and eight instead of three species correctors over a full run by at most 0.021%. The spatial error is larger and is dominated by the numerical diffusion of the first-order upwind scheme of the original campaign: at equal mesh it differs from the limited-linear scheme by 7.7–20.8% in the most sensitive case, whereas two bounded schemes differ by at most 2.8%. With limited-linear advection that case changes by 1.98% between 41,472 and 93,312 cells, its total deposit by 0.015% and its centroid by less than 1 mm, but the apparent orders are inconsistent, so we quote this change rather than a formal grid convergence index. The production cases therefore use limited-linear advection on 41,472 cells, while the follow-up sensitivities remain upwind differences on 18,432 cells. Closely competing 1 cm bins can make a reported peak temperature switch by tens of kelvin under refinement, so peaks are reported with their 3 cm smoothed maximum and centroid temperature as well.

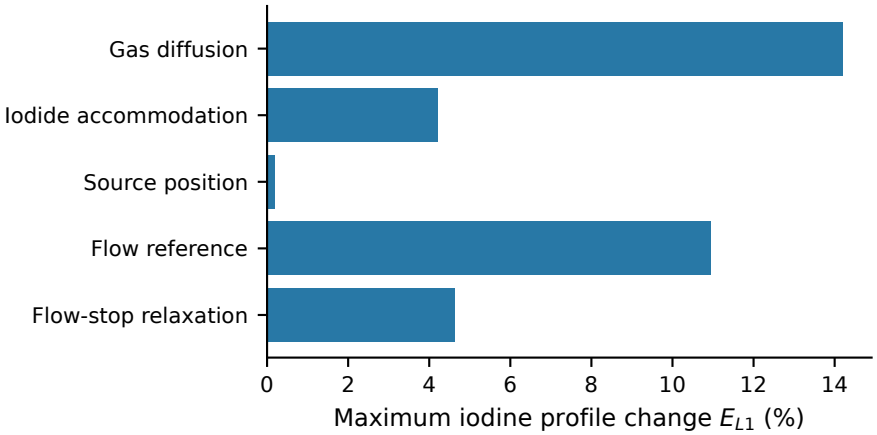


Figure 8: Maximum baseline-relative iodine profile change E_{L1} in each selected physical-input category. The flow-stop value uses an 80 s endpoint; the others use 60 s. These maxima describe the tested case set, not a general uncertainty budget or an experimentally identified parameter ranking.

Thermodynamic and source assumptions substantially affect the computed iodide partition. In the refined steel case, a $+29$ kJ/mol surrogate shift of the BiI gas Gibbs energy changes the BiI_3 iodine share from 94.58% to 2.01% (94.57% to 2.00% with upwind advection on 18,432 cells). This sensitivity survives refinement, but it does not determine which thermodynamic branch represents the experiment. Q1 agreement is conditioned on inferred deposit masses, and Q2 keeps a deposit-derived iodine source and an unmeasured metal saturation.

For silica, temperature-coordinate shifts derived from the published inputs reduce the original 4–5 cm position discrepancies to 0–1 cm. This identifies coordinate registration as a consequential uncertainty, but it does not establish a corrected experimental dataset, and the ranking of the two shifts by profile overlap reverses between the upwind and limited-linear schemes. The supplied paper resolves neither that registration nor the flow-reference conditions.

On five independent single-species PbI_2 experiments of Lobresco et al., the same wall law without any tuning gives deposition-onset temperatures with an RMS deviation of 18 K, against 35 and 35 K for the two models of that study, although the computed deposits lie 1–4 cm upstream of the measured peaks and the flow-stop transient and the non-dilute source region are not represented.

The present evidence supports a methods and input-sensitivity study. Independent predictive validation would require better-constrained source and thermodynamic inputs, consistent experimental coordinates, tabulated wall temperatures, spatially qualified profile calculations for more than one case and a treatment of the release and cooling protocol. The completed cases show where this additional evidence is needed, without hiding the original profile discrepancies.

Data and software availability

The solver and benchmark preparation scripts are maintained in the LESTO repository, https://github.com/JanKren/LESTO. The accompanying working-manuscript files contain the completed baseline, follow-up, production, scheme and mesh, and iteration-check metrics, saved axial profiles, generated tables, figure-generation code and SHA-256 provenance records. Experimental curves were digitised from Liu et al. [1]; they reconstruct published figures and are not author-supplied measurements. Figure data are adapted under the source paper's CC BY 4.0 licence. The measured profiles, wall temperatures and model curves of Lobresco et al. [6] were digitised in the same way and are adapted under that paper's CC BY 4.0 licence; the case set-up and final profiles of our comparison runs are included. Some thermodynamic coefficient files are external inputs and are not redistributed; their local identifiers and derived-table provenance are recorded with the benchmark. Reproducing the CFD calculation from coefficients requires access to those inputs. Regenerating the manuscript figures uses only the saved results and does not require them.

A Properties, thermodynamic provenance and units

This appendix lists the helium properties, molar masses, thermodynamic records and optional PbI_2–He diffusivity used in the calculations.

Table 8: Molar masses of the nine transported gases. Condensed iodide inventories use the same formula masses; the shared Pb and Bi inventories use atomic masses.

Gas	kg/mol	Gas	kg/mol	Gas	kg/mol
Pb	0.207200	Bi	0.2089804	I	0.12690447
PbI	0.33410447	Bi_2	0.4179608	I_2	0.25380894
PbI_2	0.46100894	BiI	0.33588487	BiI_3	0.58969381

A.1 Helium and transported molecular masses

The executed helium carrier uses $M_{\mathrm{He}} = 0.0040026\,\mathrm{kg\,mol^{-1}}$, $C_p = 5193\,\mathrm{J\,kg^{-1}\,K^{-1}}$ and

$$\rho = \frac{pM_{\mathrm{He}}}{RT}, \qquad \mu(T) = \frac{A_s\sqrt{T}}{1 + T_s/T}, \tag{21}$$
$$A_s = 1.4683 \times 10^{-6}, \qquad T_s = 83.40\,\mathrm{K}.$$

The numerical value of A_s gives viscosity in Pa s when T is in K. These are case inputs to the perfect-gas, constant-heat-capacity, Sutherland model, not helium properties newly fitted to measurements. Partial-pressure conversions use absolute pressure, while thermodynamic activities use $p^\circ = 10^5\,\mathrm{Pa}$. The assumed controller reference pressure, 101,325 Pa, is a separate quantity. Table 8 lists the species masses used in transport and stoichiometric conversion.

A.2 Thermodynamic records and phase treatment

Pb–I–He functions are evaluated from external NASA nine-coefficient records labelled with Gurvich source information. Bi–I functions use an external ThermoFun compilation labelled HERACLES/Barin. A common property evaluator converts all records to a consistent absolute convention, $G^\circ(T) = H^\circ(T) - TS^\circ(T)$, before forming reaction Gibbs energies. Mixing this convention with an apparent formation Gibbs energy would give inconsistent equilibrium constants. The solver tables store formation constants and phase-resolved vapour pressures as logarithms. Input-file hashes, record identifiers and table-generation scripts are recorded in the benchmark provenance.

For an iodide with one gas molecule per condensed formula unit,

$$p_{\mathrm{eq}}(T) = p^\circ \exp\left[-\frac{G_g^\circ(T) - G_c^\circ(T)}{RT}\right], \tag{22}$$

evaluated on the stable available condensed branch. The BiI_3 liquid branch is constructed at $T_{\mathrm{fus}} = 680.85\,\mathrm{K}$ from the crystalline record with $\Delta H_{\mathrm{fus}} = 31.798\,\mathrm{kJ\,mol^{-1}}$ and $C_{p,l} = 150.624\,\mathrm{J\,mol^{-1}\,K^{-1}}$, as recorded in the input-generation metadata. It is a constructed phase record, not an independently confirmed liquid thermodynamic dataset. Bi and Bi_2 share condensed Bi but have different gas chemical potentials and stoichiometry.

The BiI +29 kJ/mol variant modifies the gas record and regenerates all formation and disproportionation constants consistently. Both the baseline and the perturbed functions are hypotheses for the sensitivity calculation; the BiI source record still needs independent confirmation. The external coefficient files are not reproduced in the manuscript data package. Numerical conservation of the model derived from them cannot establish their physical correctness or complete the provenance assessment.

A.3 Optional PbI_2–He diffusion estimate

Only the PbI_2–He transport variant uses the following implemented correlation:

$$D_{\mathrm{PbI_2-He}} = \frac{1.86 \times 10^{-7} T^{3/2}\sqrt{1/4.0026 + 1/461}}{(p/101325)\sigma^2\Omega(T^*)}, \tag{23}$$
$$T^* = T/71.117857, \qquad \sigma = 4.258468, \tag{24}$$
$$\Omega(T^*) = 0.5313 - 1.2946e^{-0.9922T^*} + \frac{1.5591}{T^*} - \frac{0.1918}{(T^*)^2}. \tag{25}$$

In this numerical expression T is in K, p in Pa, σ in Å and the masses in g/mol; the resulting D is in $\mathrm{m^2\,s^{-1}}$. To reproduce the original correlation, the implementation keeps 461 g/mol rather than the more precise transport formula mass

in table 8. Its estimated collision parameters follow the original polarisability/electron-count construction recorded in the code. The other eight gases keep the constant baseline coefficient in this variant. The correlation is a screening alternative, not an experimentally validated binary diffusivity or a replacement for all missing Pb–Bi–I transport data.

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